

When Balanced Optimal Transport Breaks: A Benchmark of OT Solvers and Classical Baselines for Color Transfer under Distribution Mismatch

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Abstract

We benchmark five optimal-transport (OT) solvers from the POT library (exact EMD, entropic Sinkhorn, unbalanced OT, partial OT, and sliced OT) against three classical color-transfer baselines (Reinhard 2001, Pitié’s Iterative Distribution Transfer 2007, and the Monge–Kantorovich linear map / MKL 2007). All methods are evaluated through a single pipeline (CIELAB $\rightarrow$ K -means clustering $\rightarrow$ discrete OT $\rightarrow$ barycentric recoloring) on a Pareto plane of *palette fidelity* (Wasserstein-2 distance to the target colors) versus *structure preservation* (SSIM to the source image). Our central finding is that *balanced* OT enforces hard marginal constraints and therefore fails when the two images have a semantic color-distribution mismatch: it is forced to recolor regions that should have been left untouched. A controlled experiment (a source that is 70% green / 30% brown, a target that is 70% orange / 30% blue) shows EMD recoloring the entire green canopy to orange, whereas unbalanced OT with a small marginal-relaxation weight keeps the canopy intact (SSIM 0.31 $\rightarrow$ 0.81). Unbalanced and partial OT both recover the failure, at the cost of one extra hyperparameter that must be tuned per image pair.

1 Introduction

Color transfer (or color grading) re-colors a source image so that its color statistics match those of a target image while keeping the source content recognizable. A natural formulation treats the color distributions of the two images as probability measures and looks for a transport map that moves one onto the other at minimal cost – exactly the setting of optimal transport.

Discrete OT, in its standard (*balanced*) form, requires that *all* source mass be moved and that the transported mass reproduce the target distribution *exactly*. This is desirable when the two palettes are compatible, but it becomes a liability when they are not. If the source is a forest (dominated by greens) and the target is a red sunset sky (dominated by warm reds and a little blue), balanced OT has no feasible plan that leaves the green foliage alone: every green cluster must be shipped somewhere, and the only destinations are red or blue.

Research question. What is the failure mode of balanced OT under a semantic color-distribution mismatch, and do the relaxed OT variants (unbalanced, partial) fix it?

Contributions.

1. A unified pipeline and evaluation harness that compares eight methods (five OT solvers, three classical baselines) on identical inputs and metrics.
2. A controlled experiment that isolates the balanced-OT failure mode and quantifies it on a palette-fidelity vs. structure-preservation Pareto plane.

3. A sweep over the two relaxation knobs – the marginal-relaxation weight ρ (`reg_m`) of unbalanced OT and the transported-mass fraction m of partial OT – showing how each interpolates between “do nothing” and balanced OT.

2 Background

2.1 Discrete optimal transport

Let the source and target color distributions be represented by weighted point clouds $\{(x_i, a_i)\}_{i=1}^n$ and $\{(y_j, b_j)\}_{j=1}^m$ in CIELAB, with $a \in \mathbb{R}_+^n$, $b \in \mathbb{R}_+^m$ and $\sum_i a_i = \sum_j b_j = 1$. The ground cost is the squared Euclidean distance,

$$C_{ij} = \|x_i - y_j\|_2^2. \quad (1)$$

The Kantorovich problem seeks a transport plan $P \in \mathbb{R}_+^{n \times m}$ minimizing the total cost subject to hard marginal constraints,

$$W(a, b) = \min_{P \in U(a, b)} \langle C, P \rangle = \min_{P \in U(a, b)} \sum_{i, j} C_{ij} P_{ij}, \quad (2)$$

$$U(a, b) = \{ P \in \mathbb{R}_+^{n \times m} : P \mathbf{1}_m = a, \quad P^\top \mathbf{1}_n = b \}. \quad (3)$$

The two equality constraints in eq. (3) are what we call the *balanced* constraints: row i must send out exactly a_i , column j must receive exactly b_j . The associated 2-Wasserstein distance is

$$W_2(a, b) = \langle C, P^\star \rangle^{1/2}, \quad (4)$$

with $P^\star$ the optimal plan. Problem eq. (2) is a linear program; the POT routine [9] `ot.emd` solves it exactly.

2.2 Entropic regularization (Sinkhorn)

Adding a negative-entropy term makes the problem strongly convex and solvable by fast matrix scaling [4]:

$$P_\varepsilon^\star = \arg \min_{P \in U(a, b)} \langle C, P \rangle - \varepsilon H(P), \quad H(P) = - \sum_{i, j} P_{ij} (\log P_{ij} - 1). \quad (5)$$

As $\varepsilon \rightarrow 0$ the solution converges to the exact plan; larger ε gives a more diffuse (blurred) plan. Solved by `ot.sinkhorn`.

2.3 Unbalanced optimal transport

Unbalanced OT [5] replaces the hard marginal constraints of eq. (3) with soft Kullback–Leibler penalties, so the plan may create or destroy mass:

$$P^\star = \arg \min_{P \in \mathbb{R}_+^{n \times m}} \langle C, P \rangle + \varepsilon \text{KL}(P | ab^\top) + \rho \text{KL}(P \mathbf{1}_m | a) + \rho \text{KL}(P^\top \mathbf{1}_n | b), \quad (6)$$

where $\text{KL}(u | v) = \sum_k u_k \log(u_k/v_k) - u_k + v_k$. The weight ρ (the argument `reg_m` in `ot.unbalanced.sinkhorn_u`) controls how strictly the marginals are enforced:

- $\rho \rightarrow 0$: the penalty is cheap to violate, so the solver prefers to drop mass rather than pay a large transport cost – little is moved, the source is largely preserved.
- $\rho \rightarrow \infty$: the penalty forces $P \mathbf{1}_m \rightarrow a$ and $P^\top \mathbf{1}_n \rightarrow b$, recovering balanced OT eq. (2).

ρ has the same units as the cost matrix C .

2.4 Partial optimal transport

Partial OT [6] transports only a prescribed fraction $m \in (0, 1]$ of the total mass and is free to leave the rest in place:

$$P^* = \arg \min_{P \geq 0} \langle C, P \rangle \quad \text{s.t.} \quad P \mathbf{1}_m \leq a, \quad P^\top \mathbf{1}_n \leq b, \quad \mathbf{1}_n^\top P \mathbf{1}_m = m. \quad (7)$$

The solver keeps the most expensive $1 - m$ of the mass untransported. $m = 1$ reduces to balanced OT. Solved by `ot.partial.partial_wasserstein`.

2.5 Sliced optimal transport

Sliced OT [7] averages one-dimensional Wasserstein distances over random projections $\theta \in S^{d-1}$:

$$\text{SW}_2^2(a, b) = \int_{S^{d-1}} W_2^2(\theta_{\#}a, \theta_{\#}b) d\theta \approx \frac{1}{L} \sum_{\ell=1}^L W_2^2(\theta_\ell^\top X_s, \theta_\ell^\top X_t), \quad (8)$$

where $\theta_{\#}a$ is the pushforward of a onto the line spanned by θ and each 1-D problem has a closed-form solution by sorting. POT's `ot.sliced_wasserstein_distance` returns only the scalar eq. (8); to obtain a usable coupling we build an approximate plan as the mean of the exact 1-D couplings, $P \approx \frac{1}{L} \sum_{\ell} G_{\ell}$, with each G_{ℓ} from `ot.emd_1d` on projection θ_{ℓ} ($L = 100$).

2.6 Classical baselines

Reinhard 2001 [1]. Per-channel mean/standard-deviation matching in a decorrelated color space (here CIELAB). For each channel $c \in \{L, a, b\}$,

$$I_{\text{out}}^c = (I_{\text{in}}^c - \mu_s^c) \frac{\sigma_t^c}{\sigma_s^c} + \mu_t^c. \quad (9)$$

Iterative Distribution Transfer (Pitié 2007 [2]). Repeatedly: draw a random rotation $R \in SO(3)$, rotate both point clouds, and along each rotated axis k apply a 1-D histogram match (CDF transfer)

$$z^{(k)} \leftarrow F_{t,k}^{-1}(F_{s,k}(z^{(k)})), \quad (10)$$

with $F_{s,k}, F_{t,k}$ the marginal CDFs of the rotated source and target; then rotate back, $x \leftarrow R^\top z$. Iterating drives the full 3-D distributions together.

Monge–Kantorovich Linear / MKL (Pitié & Kokaram 2007 [3]). The closed-form W_2 -optimal map between two Gaussians $\mathcal{N}(\mu_s, \Sigma_s)$ and $\mathcal{N}(\mu_t, \Sigma_t)$ is affine,

$$T(x) = \mu_t + A(x - \mu_s), \quad A = \Sigma_s^{-1/2} (\Sigma_s^{1/2} \Sigma_t \Sigma_s^{1/2})^{1/2} \Sigma_s^{-1/2}, \quad (11)$$

with $A = A^\top \succ 0$. We take $\mu_{\bullet}, \Sigma_{\bullet}$ from all pixels of each image in CIELAB and apply eq. (11) per pixel.

3 Method

3.1 Unified pipeline

Every OT method is reduced to the same five-stage pipeline (fig. 1). Baselines skip stages 2–5 and act directly on the raw CIELAB pixels via eqs. (9) to (11).

Table 1: Dispatch table of `solve_ot`. ρ is `reg_m`.

method	POT routine	relaxation	key hyperparameters
<code>emd</code>	<code>ot.emd</code>	none (balanced)	—
<code>sinkhorn</code>	<code>ot.sinkhorn</code>	entropic	$\varepsilon = 0.05 \text{ median}(C)$
<code>unbalanced</code>	<code>ot.unbalanced.sinkhorn_unbalanced</code>	KL marginals	ε, ρ
<code>partial</code>	<code>ot.partial.partial_wasserstein</code>	mass $\leq a, b$	$m \in (0, 1]$
<code>sliced</code>	mean of <code>ot.emd_1d</code> over L slices	1-D projections	$L = 100$

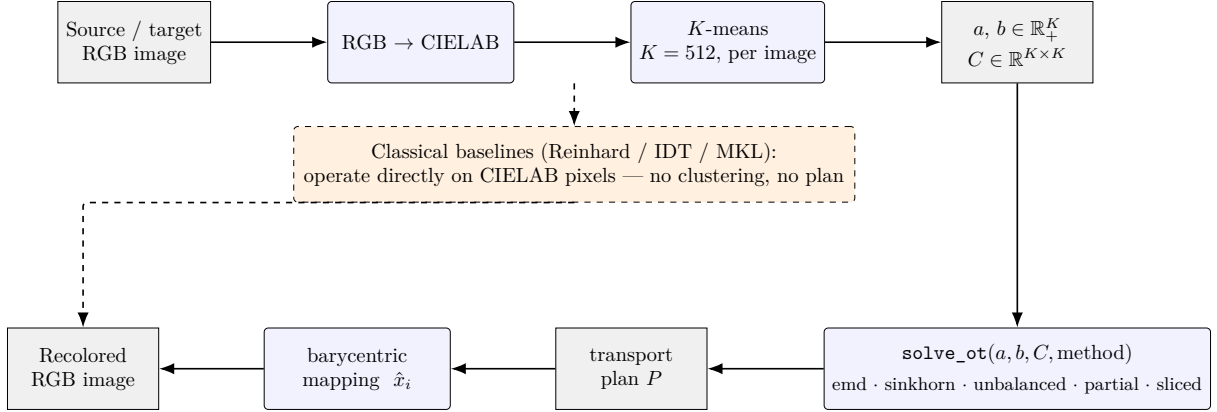


Figure 1: The unified color-transfer pipeline. OT methods follow the solid path (top row left-to-right, then bottom row right-to-left). Classical baselines follow the dashed path, skipping clustering and the transport plan.

Stage 1 – color space. Images are converted from sRGB to CIELAB, which is approximately perceptually uniform, so that Euclidean distances are a reasonable proxy for color difference.

Stage 2 – quantization. Each image is clustered independently with K -means ($K = 512$, or fewer if the image has fewer pixels). Cluster i of the source has center x_i and weight $a_i = (\# \text{ pixels in cluster } i)/N$; similarly y_j, b_j for the target. This reduces the OT problem to 512×512 .

Stage 3 – cost. $C_{ij} = \|x_i - y_j\|_2^2$ between all source/target center pairs (`ot.dist` with `metric="sqeuclidean"`).

Stage 4 – solve. `solve_ot(a, b, C, method, ...)` dispatches to the POT routine for the chosen method and returns a plan $P \in \mathbb{R}_+^{K \times K}$ (table 1).

Stage 5 – barycentric mapping. Each source cluster i is recolored to the P -weighted average of the target centers,

$$\hat{x}_i = \frac{\sum_j P_{ij} y_j}{\sum_j P_{ij}}, \quad (12)$$

and every pixel adopts the new color of its cluster. For partial and unbalanced OT the rows of P do not sum to a_i ; the undelivered mass is treated as *staying at the source color*:

$$\hat{x}_i = \frac{\sum_j P_{ij} y_j + (a_i - \sum_j P_{ij}) x_i}{a_i}, \quad (13)$$

so a cluster that ships only 30% of its mass ends up 70% its original color rather than fully recolored.

3.2 Metrics

We report a Pareto trade-off between two quantities plus a secondary check.

- **W_2 color distance** (lower is better). Empirical 2-Wasserstein distance between the *result* colors and the *target* colors in CIELAB: subsample 2000 pixels from each, form the squared-Euclidean cost, and take $\sqrt{\text{ot.emd2}(\cdot)}$. Measures palette fidelity.
- **SSIM to source** (higher is better). Structural similarity between the result and the *original source* image. Measures how much content / structure survived the recoloring.
- **Histogram intersection** (higher is better). $\sum \min(h_r, h_t)$ over a 16^3 RGB histogram; a coarse palette-overlap sanity check.

A method is good if it reaches *low* W_2 (matches the target palette) *and* keeps *high* SSIM (does not destroy the source). Balanced OT, we will see, can only do the former.

4 Experimental Setup

Images. Unless stated otherwise we use synthetic image pairs generated with NumPy so the experiment is fully reproducible and the color statistics are known exactly. Two scenarios:

- *Benchmark pair* (section 5.1): a green “foliage” source with a brown trunk band, and a red “sunset sky” target with a bright sun disc and dark foreground.
- *Mismatch pair* (section 5.2): source A is $\approx 70\%$ green / 30% brown; target B is $\approx 70\%$ orange / 30% blue. Green has no color neighbor in B .

Real photographs can be supplied on the command line instead.

Default hyperparameters. $K = 512$ (benchmark) or 64 (mismatch, where the palette is nearly piecewise-constant); Sinkhorn / unbalanced $\varepsilon = 0.05 \text{ median}(C)$ on the raw cost, or $\varepsilon = 0.02$ on the cost normalized to $[0, 1]$ in the mismatch sweep; partial $m = 0.7$ (benchmark); sliced $L = 100$ projections; K -means seed $= 0$.

Environment. Python 3, POT 0.9.7, scikit-learn 1.9, scikit-image 0.26, NumPy 2.5, Matplotlib. Timings are single-run wall-clock on CPU.

5 Results

5.1 Overall benchmark

Table 2 reports all eight methods on the benchmark pair and fig. 2 plots the Pareto plane; fig. 3 shows the images.

Table 2: Benchmark pair, $K = 512$. W_2 : distance to target colors (lower better). SSIM: similarity to source (higher better). Hist.: 16^3 histogram intersection (higher better).

method	type	$W_2 \downarrow$	SSIM $\uparrow$	Hist. $\uparrow$	time (s)
emd	OT (balanced)	4.41	0.240	0.815	0.10
sinkhorn	OT (balanced)	16.37	0.302	0.254	0.02
unbalanced	OT	45.51	0.248	0.132	0.03
partial	OT	61.38	0.358	0.570	0.13
sliced	OT	21.95	0.389	0.208	0.07
reinhard	baseline	27.17	0.519	0.033	0.02
idt	baseline	7.02	0.297	0.927	0.45
mkl	baseline	18.91	0.295	0.115	0.02

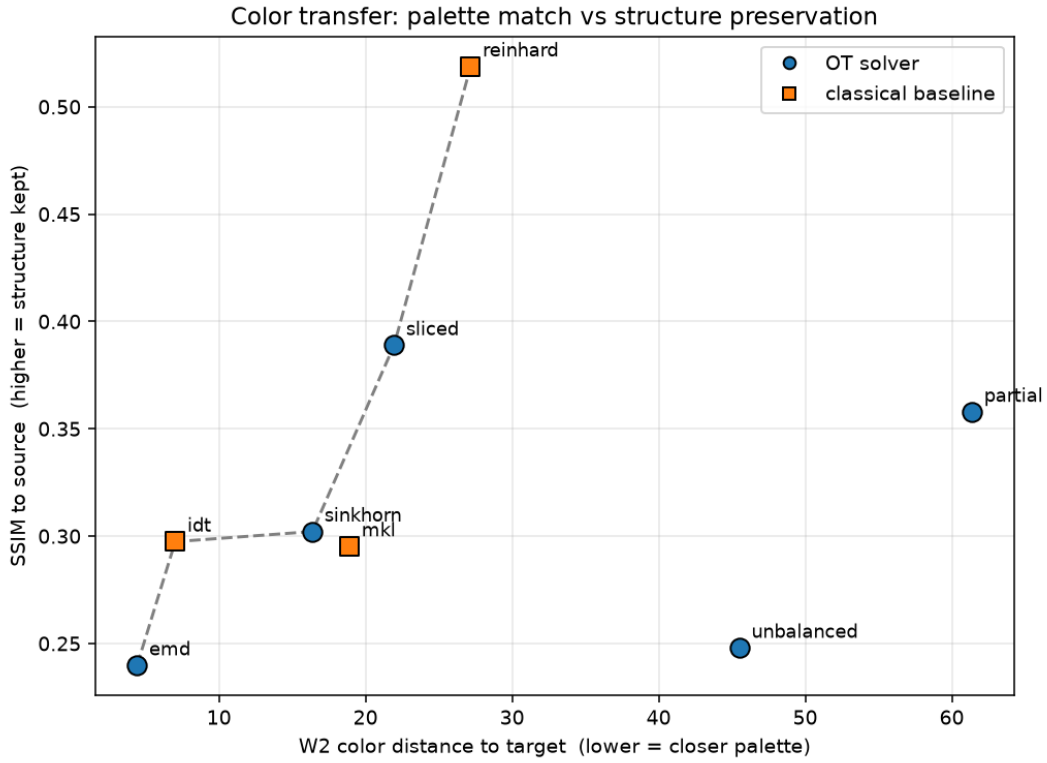


Figure 2: Pareto plane for the benchmark pair. **emd** attains the best palette match ($W_2 \approx 4$) but the worst structure score; the Pareto frontier is formed by **emd**, **idt**, **sinkhorn**, **sliced**, and **reinhard**.

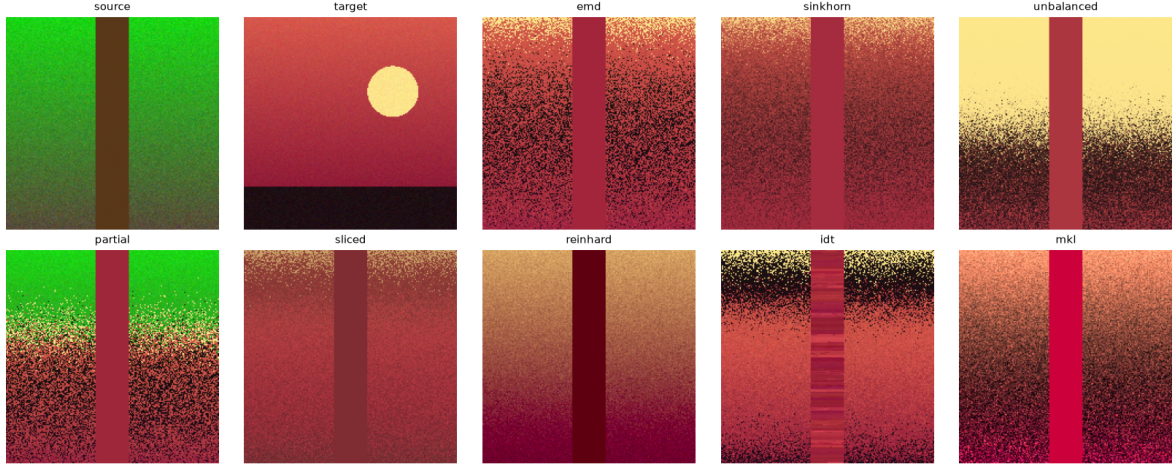


Figure 3: Benchmark pair: source, target, and the eight recolored results. Cluster-based OT (**emd**, **sinkhorn**, ...) produces salt-and-pepper speckle because neighboring pixels belong to different clusters that are mapped to different target colors; the global baselines (**reinhard**, **mkl**) are smooth continuous maps.

Observations.

1. **emd** minimizes W_2 (best palette match) but its SSIM (0.240) is essentially that of the target image itself: it reproduces the target palette by recoloring *everything*.
2. Among OT solvers, the relaxed variants (**partial**, **sliced**) keep more structure (SSIM 0.36–0.39) but at a much larger W_2 on this *compatible* pair – here there is no mismatch to exploit, so dropping mass only hurts palette fidelity.
3. The global baselines **reinhard** and **mkl** preserve structure better than cluster-based OT because they are smooth per-pixel maps; cluster-based barycentric mapping introduces quantization speckle (fig. 3).
4. **idt** is the strongest baseline here (low W_2 , high histogram intersection) as it explicitly matches the full 3-D distribution.

5.2 Distribution mismatch: the core experiment

We now use the mismatch pair, where green has no counterpart in the target. Table 3 and fig. 4 sweep EMD against unbalanced OT (varying $\rho = \text{reg_m} \in \{0.1, 1, 10\}$) and partial OT (varying $m \in \{0.5, 0.7, 0.9\}$), with the cost normalized to $[0, 1]$ so the ρ values span the useful range.

Table 3: Mismatch pair, $K = 64$, cost normalized to $[0, 1]$.

configuration	W_2 to target $\downarrow$	SSIM to source $\uparrow$
source (identity)	92.37	1.000
EMD (balanced)	12.43	0.538
unbalanced $\rho = 0.1$	87.45	0.815
unbalanced $\rho = 1$	32.13	0.583
unbalanced $\rho = 10$	10.86	0.556
partial $m = 0.5$	82.10	0.609
partial $m = 0.7$	73.54	0.585
partial $m = 0.9$	39.98	0.422

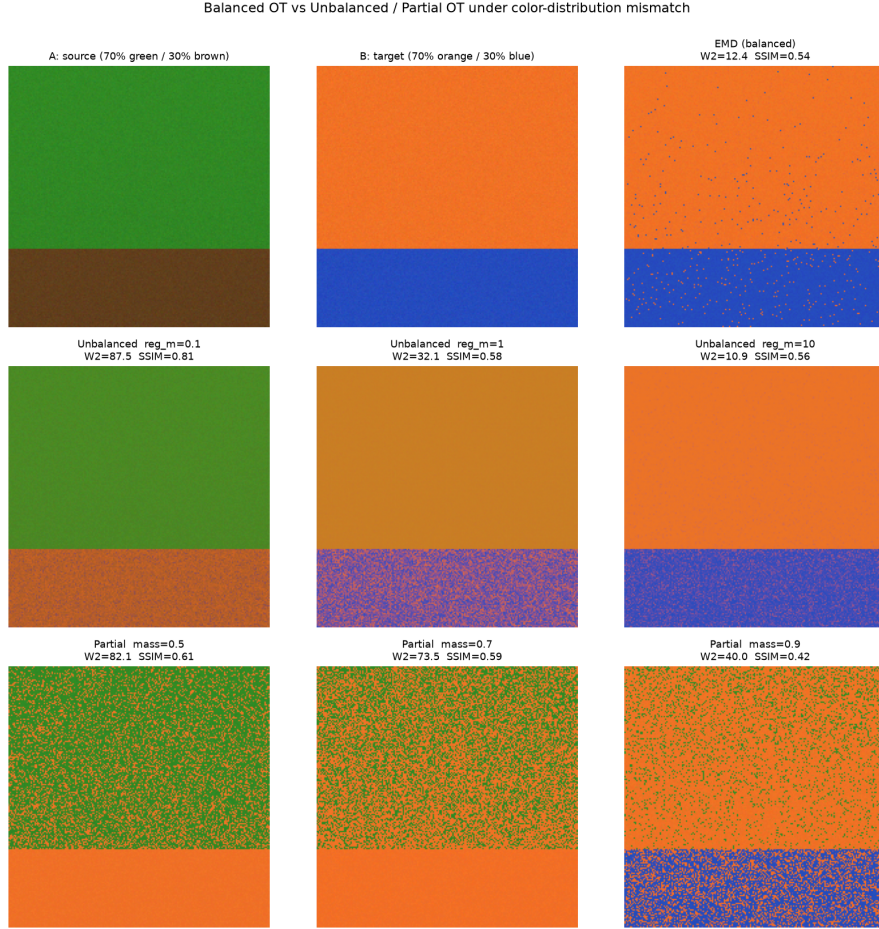


Figure 4: Mismatch pair. **Top row:** source A (70% green / 30% brown), target B (70% orange / 30% blue), and EMD. **Middle row:** unbalanced OT, $\rho \in \{0.1, 1, 10\}$. **Bottom row:** partial OT, $m \in \{0.5, 0.7, 0.9\}$. EMD recolors the green canopy fully to orange and the brown soil fully to blue; unbalanced $\rho = 0.1$ leaves the canopy green.

Observations.

1. **EMD forces implausible matches.** The green canopy becomes solid orange and the brown soil becomes solid blue – the hard marginal constraint requires the plan to deliver exactly 30% blue mass, and the only source mass available is green/brown.
2. **Unbalanced OT recovers with small ρ .** At $\rho = 0.1$ the canopy stays green (SSIM

0.815 vs. EMD’s 0.538); only the region with a plausible target shifts. As ρ grows the result moves back toward EMD ($\rho = 10$: SSIM 0.556, nearly identical to balanced).

3. **Partial OT recovers by capping mass.** At $m = 0.5$ only the cheapest half of the mass is moved and the rest is kept; $m = 0.9$ approaches balanced.
4. Both knobs are monotone interpolators between “identity” and “balanced OT”.

6 Discussion

Why balanced OT breaks. The feasible set $U(a, b)$ in eq. (3) contains *only* plans that move all source mass and exactly reproduce the target marginal. When $\text{supp}(a)$ and $\text{supp}(b)$ are far apart in color space – a semantic mismatch – every $P \in U(a, b)$ has large cost and, worse, pairs colors that should be unrelated (green $\rightarrow$ blue). The objective eq. (2) has no term that rewards “leaving a color alone”; it *cannot* express that preference.

Unbalanced vs. partial. Both introduce the missing option – to not transport some mass – but through different knobs:

- Unbalanced OT (eq. (6)) softens the marginal constraint with a penalty of tunable strength ρ . It is smooth and symmetric (it can under- *and* over-shoot a marginal), and ρ lives in cost units, which makes it scale-dependent and less interpretable.
- Partial OT (eq. (7)) caps the transported *mass fraction* m directly. m is a single, interpretable number in $[0, 1]$, but the solver – not the user – decides *which* mass to keep.

The cost: an extra hyperparameter. Neither ρ nor m has a value that is correct for every pair. Table 3 shows the SSIM/ W_2 trade-off sliding continuously as the knob turns; the “right” setting depends on how much of the source genuinely has no counterpart in the target, which is exactly the unknown quantity. In practice this needs either domain knowledge or a validation signal.

Limitations.

- Experiments use synthetic images; real photographs have smoother, higher-dimensional color manifolds.
- Cluster-based barycentric mapping (eq. (12)) causes spatial speckle (fig. 3); a spatially-regularized OT (e.g. a Laplacian penalty on the map [8]) or a larger K with post-hoc smoothing would mitigate this.
- Our sliced-OT “plan” is an average of 1-D couplings, not a true transport map; a Monge-style 1-D transport composition would be more faithful.
- Metrics are global color statistics; they do not measure local color-consistency artifacts.

7 Conclusion

Balanced optimal transport is the natural formulation of color transfer only when the two palettes are compatible. Its hard marginal constraints make it fail under a semantic color-distribution mismatch: it recolors regions it should not touch, trading almost all structural similarity for palette fidelity. Unbalanced OT and partial OT both fix this by allowing mass to be left in place, and each reduces to balanced OT in the appropriate limit ($\rho \rightarrow \infty$, $m \rightarrow 1$). The price is one extra hyperparameter that interpolates the SSIM/ W_2 trade-off and has no universally correct value.

Future work. Spatially-regularized / Laplacian OT to remove speckle; automatic selection of ρ or m from a held-out signal; evaluation on a real-photograph benchmark with mismatched semantics; a true transport map for sliced OT.

Reproducibility

```
pip install -r requirements.txt
python experiments/run_benchmark.py      # Table 1, Figures 2-3
python experiments/mismatch_demo.py     # Table 2, Figure 4
```

Source modules: `color_pipeline.py` (pipeline stages 1–3), `solvers.py` (stage 4), `barycentric.py` (stage 5), `baselines.py` (Reinhard / IDT / MKL), `metrics.py` (evaluation).

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